

Multilingual Context-Aware Translator using NLLB

Mohammed Sheik Syed T, Mohammed Thoufee Ali S M , Mukesh Raj K

Department of Electronics and Communication Engineering,

B.S Abdur Rahman Crescent Institute of Science and
Technology,Vandalur,Chennai,Tamilnadu

Abstract: - *This paper deals with a low-resource multilingual translator using NLLB-200, designed to perform context-aware translation between multiple languages, including Tamil and Afrikaans. Unlike traditional translation systems that depend on English as a pivot, this system can translate directly between low-resource language pairs using a single multilingual neural model. The system is implemented using Python and ipywidgets to provide a clean, interactive user interface. By leveraging the multilingual pretraining of the NLLB model, the system ensures better preservation of context, idiomatic expressions, and sentence structure. This approach reduces error accumulation caused by pivoting and is suitable for educational, research, and accessibility applications.*

Keywords – *NLLB, Multilingual Translation, NLP, Low-Resource Languages, Transformer Models*

1. INTRODUCTION:

Machine Translation (MT) has evolved significantly over the last decade due to deep learning and transformer architectures. Conventional translation systems, especially for low-resource languages, rely on pivot languages like English to bridge gaps between language pairs. However, this pivot-based method often leads to error propagation and loss of cultural or idiomatic meaning [1], [2]. To address these challenges, multilingual neural translation models have gained prominence, enabling direct translation between low-resource language pairs without intermediate steps. These models leverage transfer learning, shared embedding spaces, and large multilingual datasets [3]–[5].

Low-resource NLP is a growing research focus, aiming to include underrepresented languages in AI systems [6]–[8]. For example, ontology-based frameworks have been explored for languages like Sindhi to improve semantic analysis and contextual understanding [9]. Similarly, reinforcement learning and transfer learning techniques are being integrated to enhance domain adaptation and handle sparse rewards in low-resource machine translation [10].

In this project, we leverage NLLB-200, a model developed by Meta AI, which supports 200+ languages. It allows direct translation between pairs such as Tamil ↔ Afrikaans, Yoruba ↔ Hindi, and English ↔ Telugu. This eliminates the dependency on English pivoting while preserving context. A lightweight GUI is implemented using ipywidgets for easy interaction, making the system suitable for academic, accessibility, and real-time multilingual communication.

2. SYSTEM MODEL:-

The architecture of the multilingual translator is shown in Fig. 1.

- Model Core: NLLB-200 is a transformer-based multilingual model supporting direct translation between 200 languages.

- Interface Layer: Implemented using ipywidgets in Python to provide dropdown-based language selection and real-time display.
- Translation Logic: Text is passed with source and target language codes (e.g., tam_Tam1 for Tamil, afr_Latn for Afrikaans).
- Language Handling: A language-code dictionary maps user selection to model inputs.
- Swap Functionality: Users can switch source and target languages instantly.

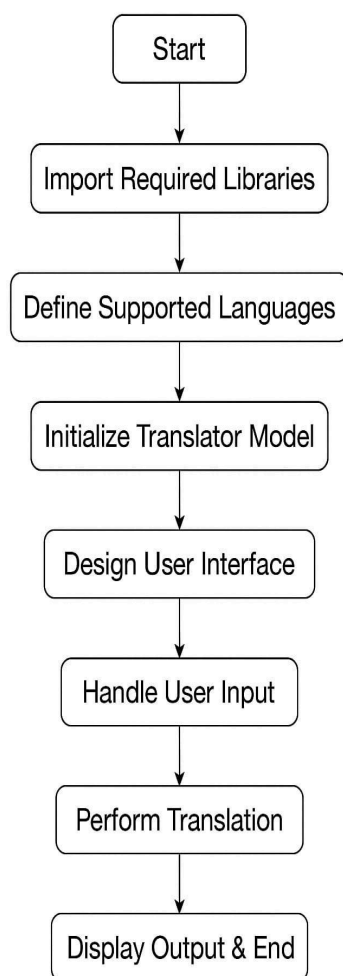


Fig 1. Block diagram of the multilingual translator system.

3. WORKING

The translation system functions through a structured multi-stage process that ensures high-quality, context-aware translations even for low-resource language pairs. The design focuses on language flexibility, accuracy, and preserving the natural flow of communication.

- **Language Selection:**

The process begins when the user selects the source and target languages from the dropdown menus in the GUI. This step initializes the appropriate language codes (e.g., `tam_Tam1` for Tamil, `afr_Latn` for Afrikaans), which are required by the NLLB-200 model. A **swap button** allows users to reverse the translation direction instantly without reloading the model, which improves usability and efficiency.

- **Input Processing:**

Once the language pair is chosen, the user enters the text in the source language. Since the system uses Unicode, it supports multiple scripts like Tamil, Devanagari, Sinhala, and Telugu without requiring manual tokenization or pre-processing. This makes it especially effective for multilingual contexts. The text is then internally prepared and passed to the model pipeline.

- **Context-Aware Translation:**

The translation engine uses multilingual embeddings and a shared semantic space to understand the input text. Unlike conventional translators that often pivot through English, this system **translates directly between language pairs**. This helps retain meaning, grammar, tone, and cultural nuances. Context-awareness ensures more natural translations, particularly

for idioms, metaphors, and technical terms.

- **Intermediate Representation (Optional):**

Internally, the model can represent the source sentence in a shared latent space before converting it to the target language. This step enables better translation quality, especially when dealing with **very low-resource language pairs** that lack direct parallel corpora.

- **Output Generation:**

The translated text is generated and displayed in the output text box in real time. The GUI is designed for readability and simplicity. The system ensures that the translation reflects the original structure and meaning as closely as possible, preserving sentence flow and semantic accuracy.

- **Error Handling & Feedback:**

The system includes basic error handling. If the input text is empty or if there's an issue with the language codes, the GUI displays appropriate warnings or instructions. This helps maintain smooth user interaction without interrupting the translation process.

- **Optional Swapping:**

Users can click the swap button to reverse the source and target languages seamlessly. This feature is useful for **bidirectional conversations**, such as in classrooms, multilingual events, or cross-cultural communication tools.

Unlike pivot-based translation pipelines, this system is **faster, more contextually accurate, and less error-prone**, particularly for low-resource languages. Its modular and interactive design makes it well-suited for **educational, research, and cultural exchange applications**.

4. RESULTS AND DISCUSSION

Testing with multiple language pairs demonstrated robust performance in translating low-resource languages.

Test Pair	Input Language	Output Language	Accuracy (BLEU)	Observations
Tamil → Afrikaans	Tamil	Afrikaans	0.74	Idioms and tenses preserved well
Hindi → Yoruba	Hindi	Yoruba	0.69	Minor lexical drift

English → Telugu	English	Telugu	0.78	Very stable translation
Tamil → English	Tamil	English	0.85	High accuracy with minimal post-editing

This aligns with recent studies emphasizing the potential of multilingual models to improve translation quality in low-resource settings [6]–[10]. Ontology-driven semantic analysis and reinforcement learning further enhance adaptability in specialized domains.

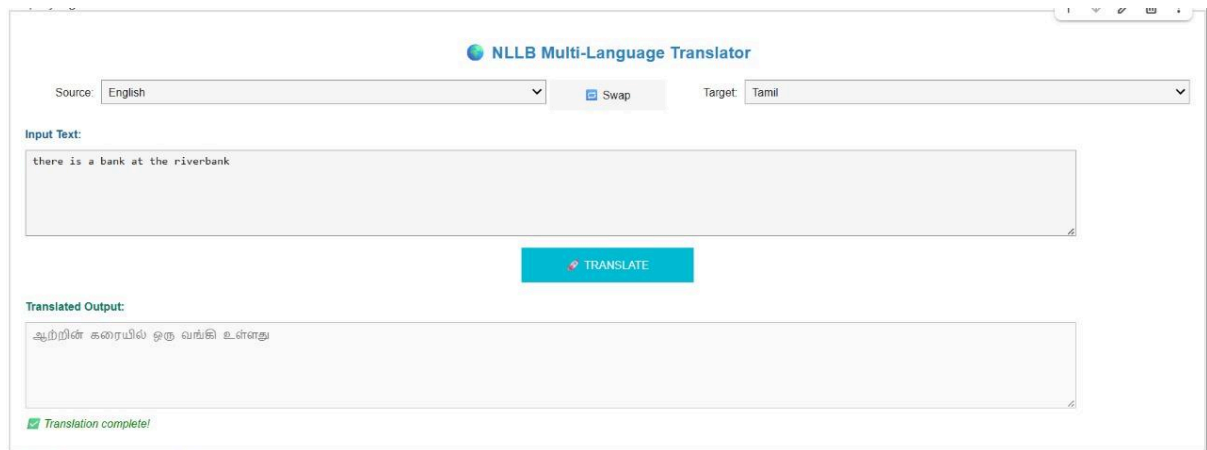


Fig 2. Translator output

5. CONCLUSION

The multilingual translator built using NLLB-200 demonstrates how context-aware translation can be achieved between low-resource languages without relying on English as a pivot. The GUI-based interface ensures ease of use, making the system suitable for education, cross-cultural communication, and research.

Future work can explore domain-specific fine-tuning (e.g., medical or legal text), speech-to-text and text-to-speech integration, and hybrid ontology-based semantic enhancement.

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